1 INVENTING THE COMPUTER

On February 15, 1946, subscribers to the *New York Times* might have been startled to discover a front-page story titled "Electronic Computer Flashes Answers, May Speed Engineering." It opened with news that "one of the war's top secrets, an amazing machine" was "heralded . . . as a tool with which to begin to rebuild scientific affairs on new foundations."[1] That evening, the new machine, ENIAC (electronic numerical integrator and computer), shown in figure 1.1, was ceremonially switched on and dedicated by Major General Gladeon M. Barnes. He represented the US Army's Ordnance Department, the owner of the new machine and sponsor of the project.

The ceremony was taking place at the University of Pennsylvania where ENIAC had been designed and built, starting in 1943. It would not be reassembled at the Ballistics Research Laboratory in nearby Maryland until 1947, where it enjoyed a long and productive career until being retired in 1955. The machine was demonstrated to reporters and dignitaries by its now-famous team of six female operators, before the visiting men joined the project engineers and managers at a lavish dinner. The 110 guests were entertained in Houston Hall, an elegant stone structure, with lobster bisque, filet mignon, ice cream, and "fancy cakes." The operators were not present, though one woman did make the guest list: Adele Goldstine, who wrote its programming manual and was married to the Army's project lead.

This was not exactly the beginning of the computer age. For one thing, ENIAC had run an elaborate set of hydrogen bomb calculations for Los Alamos starting in December of the previous year. But it was the first time anyone outside a small world of project participants and well-connected specialists with security clearance could begin to think about the possibilities opened up by machines able to carry out complex computations without human intervention while crunching numbers thousands of times faster than mechanical calculators. The press release distributed by the War Department discussed the potential applications of the new technology for nuclear physics, aerodynamics, weather prediction, oil prospecting, and aircraft design, as well as ballistics.[2]

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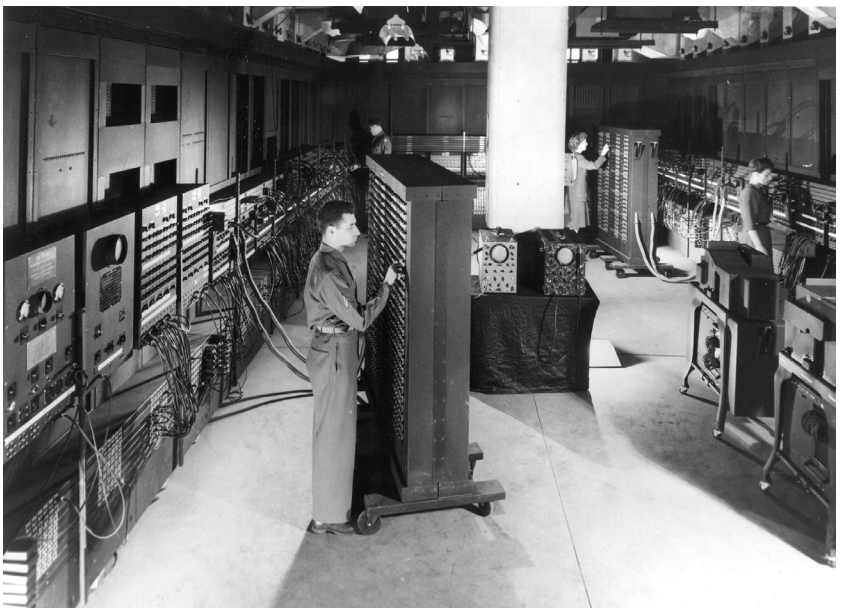

Figure 1.1
ENIAC as installed at the University of Pennsylvania, in a US Army photograph used by the *New York Times* in its 1946 report. This image defined public ideas of what an electronic computer looked like. The machine was configured by setting switches and wiring connections between its many panels, which collectively established a room within a room in which its operators and associated punched card machinery worked. Corporal Irwin Goldstein, an Army maintenance technician, is in the foreground setting data on a "portable function table" later used to hold encoded program instructions. Technician Homer Spence and two operators, Frances Bilas and Betty Jean Jennings (later Jean Bartik), work in the background.

ENIAC

ENIAC's place in computer history rests on more than being the first device to merit check marks for *electronic* and *programmable* on a comparison sheet of early machines. It fixed public impressions of what a computer looked like and what it could do. It even inspired the practice of naming early computers with five- or six-letter acronyms ending with *AC*. During a period of about five years as the only programmable electronic computer available for scientific use, ENIAC lived up to the hype by pioneering applications such as Monte Carlo simulation, numerical weather prediction, and the modeling of supersonic air flow.

Programmability

We should explain what we mean when we say that ENIAC was programmable. The idea of defining and following a program does not originate with automatic computers. Earlier meanings of *program* included a concert program, the program of study for a degree, and the programming of radio stations. In each case the program defined a sequence of

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actions over time. For example, the concert program might define three pieces of music to be performed, break the largest one into four movements, and specify an intermission. The radio station programmer allocates shows to time slots. Automatic computers similarly followed a program of mathematical operations (like addition or multiplication) rather than courses, radio shows, or musical performances. Discussion of programming a computer first appeared in the ENIAC project. By 1945 it had settled on something like its modern meaning: a computer program was a configuration that carried out the operations needed for a job. The act of creating it was called programming.[3]

ENIAC was not the first programmable computer, but it was the first to automate the job of deciding what to do next after a sequence of operations finished. Consider the context in which it appeared. In the summer of 1944, just as Harvard's Mark I was publicly unveiled, the Navy ordered Grace Murray Hopper to the Computation Lab to assist Howard Aiken with programming it. Hopper had been an assistant professor of mathematics at Vassar College and had taken leave to attend the Navy's Midshipmen's School. According to Hopper, she had just earned her one-and-one-half stripes when she reported to the lab at Harvard. There, Howard Aiken,

> . . . a large object, with three stripes . . . waved his hand and said: "That's a computing machine." I said, "Yes, Sir." What else could I say? He said he would like to have me compute the coefficients of the arc tangent series, for Thursday. Again, what could I say? "Yes, Sir." I did not know what on earth was happening, but that was my meeting with Howard Hathaway Aiken.[4]

Thus began the practice of computer programming in the United States. Hopper wrote out the sequence of codes for that job, and later the codes for more complex mathematical tasks. One of the first was a design problem for James Baker, a Harvard Fellow responsible for the lenses used for top-secret US spy cameras.

Early relay computers relied on human operators to step in when decisions had to be made. The Harvard Mark 1, for example, spent most of its time running loops, created by gluing the ends of the control tape together. It could automatically stop work after finishing that stage in the computation, leaving its operator to figure out which strip of program tape to mount next. As the Mark 1 took six seconds to carry out a multiplication, and a full minute to execute built-in sequences for logarithmic and trigonometric operations, this reliance on human intervention for what we would now call program branching did not initially seem to be a major performance limitation.

ENIAC Design and Construction

During the Great Depression, John Mauchly, the ENIAC project's instigator, was a PhD physicist struggling to carry out meaningful research under a heavy teaching load at a small college. One of his interests was the use of electronics for counting. As war

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loomed for America, the government sponsored a special summer school in electronics at the University of Pennsylvania. This crash training drew on the Moore School's expertise in the new technology and ties to local industry. Mauchly distinguished himself, which led to a professional step up to a faculty position.

Mauchly learned that hundreds of women were employed at the Moore School to compute firing tables, under contract to the US Army's Ballistics Research Laboratory. When firing shells and mortars, the gun operator would begin with the estimated distance of the target. He would then refer to a book of mathematical tables specific to that combination of gun and munition to find the correct angle. The answer varied depending on change in elevation, temperature, and wind speed.[5] Expressing the motion of a shell as a differential equation was straightforward. Test firing each gun a few times at Aberdeen Proving Ground yielded parameters to plug into this standard equation. The problem was that, like most other real-world differential equations, this one could not be tackled with the methods taught in calculus classes. Because of atmospheric drag the (human) computers had to solve the equation numerically, breaking the shell's flight into several thousand intervals to calculate changes in its speed and position from each instant to the next. The simulated trajectory provided just one row of the firing table, giving the shell's range for one angle of elevation. Producing the entire table by hand took months of work. Hard as the computers worked, their backlog of work grew ever larger. New guns were being shipped to Europe without the tables needed to operate them.

That backlog gave Mauchly the perfect excuse to justify development of an electronic computing device. Since the 1980s, a program so compelling that users acquire an entire computer system to run it has been called a killer application. This term is particularly appropriate to ENIAC, as the aim was literally to kill more efficiently. His chief collaborator was Presper Eckert, a master's degree engineering student who stayed at the Moore School to help run its wartime projects.

Because ENIAC was both electronic and general purpose, its designers faced a unique challenge. In Mauchly's words, "Calculations can be performed at high speed only if instructions are supplied at high speed."[6] That required a control method faster than paper tape. It also meant avoiding frequent stops for human intervention. For example, ENIAC might need to repeat a numerical procedure until a sufficiently precise result was obtained and then automatically move on to the next step of the calculation. Mauchly sketched out several possible mechanisms to select automatically between different preset courses of action depending on the values ENIAC had already calculated. Computer scientists call this conditional branching and view it as a defining feature of the modern computer.[7]

ENIAC's unique control method was based on a careful analysis of the firing table problem. Early in the project, Arthur Burks, one its key designers, worked with Adele

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Goldstine, a mathematician who had been overseeing the women calculating firing tables. They produced a draft configuration to tackle the problem and devised methods to diagram the sequence of mathematical operations needed and the physical setup of ENIAC that would generate them. This was implemented by connecting wires to carry "program" pulses from one unit to another. When an addition was followed by a division, a wire ran from an output terminal of an adding and storing unit (accumulator) to an input terminal on the divider. The exact operation triggered by the pulse was determined by switch settings. After the pulse primed the divider to expect input, a separate network of cables fed it the numbers to be divided. Altogether, ENIAC was not so much a single computer as a kit of forty modules from which a different computer was constructed for each problem.

While Eckert and Mauchly are the names usually attached to ENIAC, dozens of people worked to create the machine. A team of about six design engineers was assisted by a secretary, draftswomen, and model builders. Mathematical experts at Penn and the Ballistics Research Laboratory helped to figure out what capabilities the machine would need. University administrators and procurement specialists scrambled to obtain obscure components. By mid-1944, as work shifted from prototyping to full-scale production, around forty people were employed to assemble ENIAC.

ENIAC in Use

It took ENIAC 30 seconds to simulate a trajectory, potentially tracing the shell's journey faster than the shell itself could travel. The war was over by the time ENIAC was finished, so firing table preparation became less urgent. In the end, ENIAC spent something like 15 percent of its production time calculating them and the rest on other, varied jobs, including many to aid the Los Alamos and Argonne laboratories in the development of nuclear weapons and reactors.

ENIAC was also a workplace of around two thousand square feet. Its panels were arranged in a U shape, working like a set of room dividers to enclose an inner space in which its operators worked. Its original home in the Moore School's basement was a little shabby, but the Ballistics Research Laboratory built it a smart and modern home with the only air conditioning unit on the Proving Ground to dissipate heat from its 18,000 or so vacuum tubes. Staff moved their desks inside ENIAC to escape from the summer heat and humidity. Groups of visitors, from cadets to President Truman, were frequently brought to marvel at ENIAC, establishing a tradition of computer rooms as organizational showpieces that lasted through the 1960s.[8]

Today, ENIAC is most often remembered as a machine programmed by women. When in use, like other early computers it needed frequent human attention. ENIAC was typically staffed by two operators, joined frequently by a supervisor or by the scientists for whom the calculation was being carried out. Six women hired in mid-1945

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were its operators during an initial period of use in Philadelphia from December 1945 to December 1946. They have become famous as the "women of ENIAC" and are often called the first computer programmers, though remembering them only as programmers doesn't reflect the full range of their work.

Data went in and out of ENIAC on punched cards: small rectangles of cardboard each able to store 80 digits as a pattern of holes. The women spent much of their time punching input data onto cards and running output cards through an IBM tabulating machine to print their contents. Bigger jobs produced thousands of intermediate cards, which they had to process manually using other IBM machines before feeding them back in. They followed configuration diagrams to reconfigure the computer's wires and switches when changing between problems. Because of the expertise they built up, the women often helped scientists and engineers to formulate their problems as sequences of ENIAC operations and convert those sequences to diagrams. On later computers these tasks were usually handled by separate teams of key punch women, computer operators, and programmers. Even ENIAC was eventually operated around the clock by three shifts of operators after it proved itself at Ballistics Research Laboratory.

THE EDVAC APPROACH

By mid-1944, more than a year before ENIAC was working, its creators were already working on ideas for a successor. They recognized that switching ENIAC between jobs would require a lengthy and error-prone period of reconfiguration and that its distributed control method imposed a confusing mass of bottlenecks limiting, for example, the number of branches that could be included in a program.

The proposal to build EDVAC (electronic discrete variable automatic computer) was approved by the government in August. It centered on a new kind of memory proposed by Eckert, the delay line.[9] ENIAC used twenty-eight vacuum tubes to hold each decimal digit. That approach would not scale far. It took years of engineering frustrations to make delay line memory work reliably, but the idea was simple and compelling. Pulses representing several hundred digits moved through a fluid-filled tube. Signals received at one end were immediately retransmitted at the other end, so that the same sequence was cycling constantly. Whenever a number reached the end of the tube it was available to be copied to the computer's processor if needed.

Von Neumann's *First Draft*

How to design a computer around this new storage system? In the summer of 1944, shortly before the formal proposal for EDVAC was written, the ENIAC team welcomed a new collaborator. The famous mathematician John von Neumann was sufficiently

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intrigued by word of what was going on at the Moore School to have himself brought onto the project, despite his many other urgent collaborations, including the design of the atomic bomb. Engineering work on the new machine was delayed when ENIAC's completion slipped by more than a year, monopolizing the school's resources. During this period, the Moore School team discussed with von Neumann a variety of ideas for how to structure EDVAC. He pulled some of the proposals together into a coherent whole during a trip to Los Alamos, adding a control scheme with a standard instruction format and a simple but flexible set of instructions.

Von Neumann's *First Draft of a Report on the EDVAC* described logical structures rather than the specifics of hardware. One of its most novel features was that, as the team had decided by September 1944, coded instructions were stored in the same storage devices used to hold data. The ideas contained in the *First Draft* are frequently summed up in a single phrase: "the stored program concept." In April 1945, Goldstine circulated von Neumann's text within the Moore School.[10] It is often called the founding document of modern computing. A lightly revised version, dated June 30, went out to about two dozen people interested in electronic computing, most of them already connected to the ENIAC project. As it took more than a year for anyone to get around to filing a patent on ENIAC, this disclosure of the new architecture put the modern computer into the public domain.

The idea of loading a program into main memory was important and set EDVAC apart from existing computer designs. However, following work done by Haigh in collaboration with Mark Priestley and Crispin Rope, we prefer to separate the enormous influence of EDVAC into three clusters of ideas (or *paradigms*).[11]

The first of these, the EDVAC hardware paradigm, specified an all-electronic machine with a large high-speed memory using binary number storage (figure 1.2). Eckert and Mauchly were committed to this before von Neumann arrived. From their viewpoint, everything else in the report was a working through of the implications of this new technology, and they resented the credit that von Neumann received for its contents.

The second was the von Neumann architecture paradigm. Influenced by the emerging field of cybernetics, which asserted the equivalence of brains and mechanical control mechanisms, von Neumann borrowed biological language to describe EDVAC's switching circuits as *neurons*.[12] He broke its structure into *organs*, including a large delay line *memory* for programs and data (the only biological term that stuck). Storage and arithmetic was binary, using what would soon be called bits (a contraction of *binary digits*) to encode information. Each 32-bit chunk of memory (soon to be called a word) was referenced with an address number. The program to be run would be loaded from R, the *recording medium*, into the main memory, M, before being executed. All data being worked on would also be copied into memory. This differed

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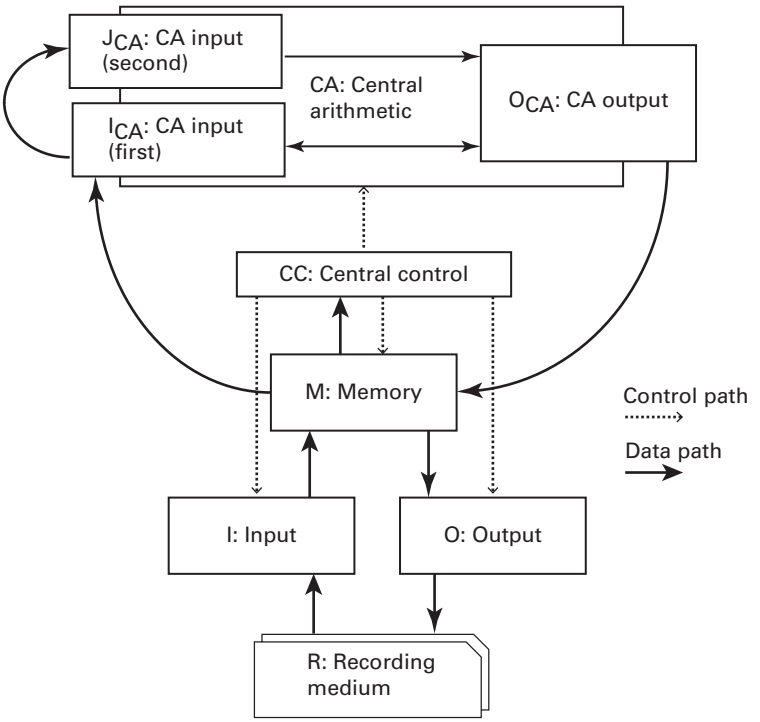

Figure 1.2

The logical architecture of EDVAC as described in the *First Draft*. Reproduced from Thomas Haigh, Mark Priestley, and Crispin Rope, *ENIAC in Action* (Cambridge, MA: MIT Press, 2016, p. 145).

from the relay calculators, which read instructions one at a time from tape, executing each immediately. ENIAC held many data pathways between units, and put adding circuits into every storage unit. EDVAC, in contrast, would have just one connection each way between memory and the *central arithmetic* organ where all arithmetic was centralized. Issues such as the number of adding units and connections between them and memory had been debated during von Neumann's meetings with the Moore School team, but in the report he selected a set of design options and combined them into an elegant and coherent architecture.

The third cluster of ideas was a system of instruction codes: the modern code paradigm. The flow of instructions and data mirrored the way humans performed scientific calculations as a series of mathematical operations, using mechanical calculators, books of tables, and pencil and paper. Even Eckert and Mauchly credited von Neumann with devising the proposed instruction code. It represented each instruction with an operation code, usually followed by parameters or an address. Because the architecture was so simple, even adding two numbers and storing the result took at least four instructions: reading the first number from M into the location marked I_{CA} in the diagram,

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reading the second number into I_{CA} (which automatically shifted the first into J_{CA}), adding them (which stored the result in O_{CA}), and copying this value back to M (overwriting the second number). In later terminology the special storage circuits, such as $I_{CA,}$ were called *registers*. The one used, like $O_{CA,}$ as the destination for mathematical operations was called the *accumulator*.

Each of these four instructions took several steps to carry out. The basic cycle of an EDVAC-like computer is to fetch the next instruction code from memory (in this case copying it into the *central control* organ), decode that instruction, and execute it. A special circuit, the program counter, holds the number of the address from which the next instruction should be fetched. The program counter is automatically incremented by one every time an instruction is read (hence the term *counter*), so that the next instruction is fetched from the next memory location. Manipulating the number stored in the counter causes a *branch* to a different part of the program code.

Most computers today harness several processor cores running in parallel, but the concept of processing a stream of instructions from an addressable memory remains the most lasting of all the *First Draft*'s contributions. Computer scientist Alan Perlis remarked that "sometimes I think the only universal in the computing field is the fetch-execute cycle."[13]

Early EDVAC-Like Computers

All three paradigms were hugely influential in the early days of computing. Recognizing the eagerness of would-be computer builders to see ENIAC and learn more about the ideas contained in the *First Draft*, in the summer of 1946 the Moore School and the US military co-sponsored a course on the "Theory and Techniques for Design of Electronic Digital Computers."[14] The same year, von Neumann, now overseeing a computer project at the Institute for Advanced Study in Princeton, began work with Arthur Burks and Herman Goldstine on an influential series of reports to develop ideas on computer architecture, instruction sets, and programming.[15] Between them, the Moore School lectures and the IAS reports firmly established the new approach to computer design.

The new computers had simple architectures, but making them work took years of engineering frustration. Early computers had to transmit digital pulses lasting perhaps one hundred thousandth of a second through miles of wire with complete reliability. They relied on digital logic circuits of huge complexity. They included thousands of vacuum tubes, which under normal use would be expected to fail often enough to make any computer useless. Building one meant soldering hundreds of thousands of electrical joints. But the biggest challenge was producing a stable memory able to hold tens of thousands of bits stable long enough to run a program. Delay lines were simple in theory but treacherous in practice. The Moore School tried building a delay line memory for

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ENIAC, under contract to the Ballistics Research Laboratory. After two years it was installed, but it never worked well enough to be useful and the laboratory returned it.

As a stopgap, ENIAC was reconfigured in March and April 1948 to run modern code (in the sense previously defined) entered by turning switches on read-only memory panels. It was used in that mode for the rest of its career. Adele Goldstine sketched out the original plan for the conversion and worked to refine it with Jean Bartik, one of the original operators who had been rehired in 1947 to run a contract programming group (the first of its kind) to develop ENIAC applications. Klara von Neumann, wife of John, did the detailed coding for the first modern program executed: a complex Monte Carlo simulation run for Los Alamos to observe virtually and experiment with the chain reaction inside an exploding nuclear weapon. The contributions made by those women, as well as the mostly female team that wired and assembled the machine, make it misleading to talk about just six people as "*the* women of ENIAC."[16]

A few months later, a team at the University of Manchester, England, debugged its writable electronic memory with the help of a tiny test computer. This was a stepping stone to a useful EDVAC-like computer, now called the Manchester Mark 1. Even when they could be made to work, delay lines were rather slow, as the computer had to wait for the bits of interest to take their turn cycling through the sensing equipment at the end of the tube. The Manchester group had a faster solution. The device popularly known as the Williams tube, after engineer Freddy Williams, stored bits as charges on a cathode ray tube (CRT) similar to those used in televisions and radar sets of that period. This produced a pattern of visible dots on the tube. By adding a mechanism to read charged spots as well as write them, a single tube could be used to store two thousand bits. The challenge was to read them reliably and to constantly write them back to the screen before they faded away. This technology was licensed by other computer builders, including the Institute for Advanced Study and IBM.[17]

In the spring of 1949, before the full Manchester computer was ready, Maurice Wilkes of Cambridge University managed to get his EDSAC operational. This made it the first full-scale computer to implement all three of the EDVAC paradigms. EDSAC became the basis of a scientific computing service, solving problems in biology, chemistry, and radio astronomy as well as physics and mathematics. One of its beneficiaries, John Kendrew, won a Nobel Prize after EDSAC was programmed to aid in the otherwise infeasible task of deducing the three-dimensional structure of proteins from the scattering of X-rays.[18]

Variations on a Theme

Each computer project launched in the 1940s was an experiment, and designers tried out many variations on the EDVAC theme. Even computers modeled on von Neumann's slightly later IAS design, built at places like Los Alamos, the Bureau of Standards, and the RAND Corporation, diverged by, for example, using different memory technologies.

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Successful ideas were widely copied. For example, many programs worked on data held in a matrix structure (essentially a table). The programmer defined a loop to repeat a sequence of operations for each cell in the matrix. Each time it looped, the code had to act on data held in a different memory location. In most computers of the 1940s this was accomplished by having the code modify the instruction held in memory, an idea introduced in the *First Draft*. Beginning with the University of Manchester's Mark 1 computer in 1949, designers added to the processor what would later be called an index register to simplify work with data structures. The instructions stayed the same. Instead, the program updated an offset value in the index register, which was added to the *base address* held in the instruction to determine the actual memory location acted on.[19]

Storing programs and data in addressable memory was a hallmark of the EDVAC approach. A single memory location was a *word* of memory. But computer designers made different choices about how large each word should be and how instructions should be encoded within it. The original EDVAC design called for thirty-two bits in each word. Early computers used word lengths between seventeen (EDSAC) and forty (the IAS computer and the Manchester Mark 1) bits. A larger word size made working with bigger numbers convenient and efficient but demanded more hardware and was wasteful if most numbers and instructions did not fill an entire word. Maurice Wilkes chose a short word length for EDSAC to efficiently store instructions but included hardware support for numbers that filled two words, a practice that later became standard.

One factor influencing word length was the maximum number of addresses specified in a single instruction. Computer architects saw this as a crucial design feature to the extent that they used *one address*, *three address*, and so on as a taxonomy to define classes of computer. The EDVAC instruction set defined in the *First Draft* specified either a source or a destination address when reading or writing main memory, making it a one-address machine. Its arithmetic operations functioned on numbers that had already been loaded into special registers and so were *zero-address* instructions. But EDVAC, as eventually built, had grown into a *four-address* machine. Its addition instruction included memory locations for both numbers to be added and a third for the location in which to store the result. The fourth address specified the location of the next instruction to be read. The example we gave previously, using four instructions to add two numbers and store the result, would take just one instruction in the final machine.

Almost every computer operation involved moving data backward and forward between registers, arithmetic units, and main memory. One crucial engineering decision was whether to move all the bits in a word sequentially on a single wire or send them together along a set of parallel wires. This was the original meaning of *serial* and *parallel* in computer design. EDSAC was serial, a popular choice for machines with delay line memories, which delivered data one bit at a time. Serial transmission of bits

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was slower but simpler. Most parallel-word computers used Williams tubes, which could retrieve all the bits in a word of memory simultaneously.[20]

You may have noticed that we have not yet mentioned two of the most famous figures in the history of computing: Charles Babbage and Alan Turing. Babbage's efforts a hundred years earlier to build a mechanical computer were remarkable but had no direct influence on work in the 1940s; the ENIAC team didn't know about them and even Howard Aiken, who helped to revive Babbage's reputation as a computer pioneer, designed his computer in ignorance of the details of Babbage's work.[21]

Alan Turing's conceptual work on computability carried out in the 1930s was later foundational to the development of theoretical computer science. While von Neumann was aware of, and intrigued by, Turing's concept of a "universal machine," we see no evidence that it shaped his design for EDVAC.[22] In 1946, however, Turing proposed a radical reinterpretation of the EDVAC approach for the computer project at Britain's National Physical Laboratory. His ACE design coded every operation as a transfer between delay lines. Some transfer destinations carried out special operations, such as adding the number it received. The architecture was commercialized on some early British machines, which were simple and fast but very hard to program because programmers had to track exactly when each value they wanted to work on would be emerging from a delay line and time their code accordingly.[23]

Programming Tools

General purpose computers can do many things. The disadvantage to that flexibility is that getting any particular task done takes minutely detailed programming. It did not take Grace Hopper long to realize that reusing pieces of Mark 1 code for new problems could speed this work. Her group built up a paper tape library of standard sequences, called subroutines, for routine operations such as calculating logarithms or converting numbers between decimal and binary format. In Germany, Konrad Zuse envisioned a "Plan Preparation Machine" (*Planfertigungsgeräte*).[24] It would punch control tapes for his Z4 computer on the basis of commands entered by a user, automatically checking that their syntax was correct.

The arrival of EDVAC-like computers opened up new possibilities for automating program preparation. The computer itself could be programmed to handle the chores involved in reusing code, such as renumbering memory addresses within each subroutine according to its eventual position in memory. These new tools were called assemblers because they assembled subroutines and new code into a single executable program. Assemblers quickly picked up another function. Humans found it easier to refer to instructions by short abbreviations, called mnemonics. For example, LOAD might be the mnemonic for the instruction to copy a number from main memory into a processor

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register. The list of instruction mnemonics and parameters was called assembly language. The assembler translated each line into the corresponding numerical instruction that the computer could execute.

Of all the 1940s computers, EDSAC had the most convenient programming system. Every time the machine was reset, code wired into read-only memory was automatically triggered to read an instruction tape, translating mnemonics on the fly and loading the results into memory. David Wheeler developed an elegant and influential way of calling subroutines, so that the computer could easily jump back to what it was doing previously when the subroutine finished.[25] EDSAC users built a robust library of subroutine tapes and published their code in the first textbook on computer programming.[26]

The general arrival of *symbolic* and *macro* assemblers in the 1950s made assembly coding less tedious. As instructions were added and removed from a program, the memory locations holding variables or subroutines would change. Symbolic assemblers let programmers use labels rather than numbers to specify addresses, which eliminated the need to edit the code every time locations changed. Unlike a regular mnemonic, which produced a single machine instruction, a macro instruction expanded into an entire block of code. For example, macro instructions made it easier to retrieve information from tape files.

The first symbolic assembler was written by Nathaniel Rochester in 1949 as part of an internal IBM project to experiment with the EDVAC approach. The test assembly lashed the arithmetic from IBM's new 604 Electronic Calculating Punch, its first commercial electronic calculator, to a variety of custom components including a new control unit and larger drum memory. The 604 already had a built-in control mechanism via simple programs wired on its plug board. The team called the EDVAC-style instructions held in the drum memory its stored program, to distinguish it from the plug board controls. As this term was taken up by others, its meaning gradually broadened until "stored program" became short-hand for the whole cluster of new ideas about computer architecture introduced in the *First Draft*.[27]

The Association for Computing Machinery

As interest in electronic computers spread, a new organization, the Association for Computing Machinery, was founded in 1947 for the exchange of information and ideas between those working in the new field. Its first members had come together in a series of special meetings and symposia, but they recognized the benefits that an association, with a regular conference series, could bring.[28] The indefatigable Edmund Berkeley served as its founding secretary. The new association developed slowly at first, beginning publication of a journal in 1954.

The ACM eventually grew into a society of around a hundred thousand members, with a host of special interest groups, publications, and conferences. While our focus

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here is on computing practice and technology rather than computing research, we will be noting many of the winners of its A.M. Turing Award, first presented in 1966. This is the premier prize in computer science, promoted by ACM as the equivalent of a Nobel Prize. It marks contributions that the computing research community recognizes for exceptional intellectual merit or, in a few cases, undeniable practical influence.

Computer science did not exist as a discipline in the 1950s, so those researching the development and use of computers came from a variety of disciplinary backgrounds. Engineering societies were also important venues, particularly the Institute for Radio Engineers and the American Institute of Electrical Engineers. In 1963 those merged to create the Institute of Electrical and Electronics Engineers (IEEE), whose Computer Society developed as the other main organization in the field.

THE COMMERCIALIZATION OF COMPUTING

"[Y]ou . . . fellows ought to go back and change your program entirely, stop this . . . foolishness with Eckert and Mauchly." That's what Howard Aiken told Edward Cannon of the US National Bureau of Standards in 1948, as part of a committee recommending against supporting their proposal to make and sell electronic computers.[29] Eckert and Mauchly had a vision, not only of how to design and build a computer but also of how a society might benefit from large numbers of them. In Aiken's view, a commercial market would never develop: in the United States there was a need for perhaps five or six such machines, but no more.[30] Aiken was quickly proven wrong. Thousands of computers were sold in the 1950s.

Eckert and Mauchly's drive to build computers was matched by an equal but opposite drive by the University of Pennsylvania to banish commercial interests from the academy. Irwin Travis, an administrator at the Moore School, demanded that members of its staff sign a form relinquishing ownership of inventions made during their work there. Eckert and Mauchly refused to sign. They resigned effective March 31, 1946. Work on the EDVAC was turned over to a succession of other engineers who took five more years to finish it.[31]

The First Computer Start-Up

Eckert and Mauchly could have found work elsewhere, but instead they chose the risky course of founding their own company. They formed a partnership, the Electronic Control Company, in 1946; in December 1948 they incorporated as the Eckert-Mauchly Computer Corporation. Beyond the engineering problems of designing and building a computer and its associated tape drives, memory units, and input-output equipment loomed the bigger problem of raising capital. Their solution was to raise

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money by preselling the computers. The US Census Bureau was interested, which led to the investigation undertaken by Aiken and his colleagues. The frantic search for funds even took them to American Totalisator Company, who wanted a computer to calculate betting odds at race tracks. On January 12, 1948, John Mauchly wrote a memorandum to his fellow staff at the Eckert-Mauchly Computer Corporation in which he listed a total of twenty-two industries, government agencies, and other institutions that he had contacted.[32] Among them were

> Prudential. [Edmund C.] Berkeley . . . says that considering the number of persons at Prudential who have now expressed themselves in favor of obtaining electronic equipment, he believes there will be no difficulty in getting an order for one Univac.
>
> Oak Ridge . . . it was almost 100% certain that their purchase order would be approved by Army . . .
>
> Army Map Service . . . Army Map Service has taken an interest in Univac equipment.
>
> Bureau of Aeronautics . . . we could possibly obtain a contract . . .
>
> The Metropolitan Insurance Company has a large problem involving a total file of 18,000,000 policies with 2,000,000 changes per week. There are about 20 digits of information for each policy. It appears that this is a natural application for the Univac . . . it would be worthwhile to follow it up . . .
>
> Aircraft Companies. A number of aircraft companies are good prospects . . . there is no doubt that such companies could use Univac equipment. We have had brief contact with Hughes Aircraft, Glen L. Martin, United Aircraft, North American Aviation, and have been told that Grumman goes in for some rather fancy calculations.

The undercapitalized company struggled to turn these prospects into orders. That wasn't enough to save the Eckert-Mauchly Computer Company, even though its main product, the Univac, established the marketplace for computers. It became a division of the business-machines company Remington Rand. But all these organizations, and many more, would soon find compelling reasons to purchase or lease electronic digital computers, if not from Eckert-Mauchly then from someone else.

Eckert and Mauchly, with the help of about a dozen technical employees of their division, designed and built the Univac in a modest factory at 3747 Ridge Avenue in Philadelphia (see figure 1.3). It used four bits to code each decimal digit. In its central processor, four general-purpose accumulators carried out arithmetic. A word was 45 bits long; representing 11 decimal digits plus a sign, or six alphabetic characters. The Univac's clock ran at 2.25 MHz. It could perform about 465 multiplications per second, not a spectacular advance on ENIAC, but its tape system and larger memory made it a much faster machine overall. Delay lines stored one thousand words as acoustic pulses in tubes of mercury, while magnetic tape units stored up to one million characters on each reel of 1/2-inch metal tape. Its central processor contained over

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Figure 1.3
The staff of the Eckert-Mauchly Computer Company gather in 1948 at their Philadelphia factory, in front of their first computer, the custom BINAC created for military aircraft developer Northrop. Front row (left to right): J. Presper Eckert (cofounder and ENIAC chief engineer), Frazier Welsh, James Wiener, Bradford Sheppard, and John Mauchly (cofounder and ENIAC project initiator). Back row: Albert Auerbach, Jean Bartik (former ENIAC operator and programming manager), Marvin Jacoby, John Sims, Louis Wilson, Robert Shaw (former ENIAC engineer), and Gerald Smoliar. Courtesy Unisys Corp.

five thousand tubes installed in cabinets arranged in a 10′ × 14′ rectangle. Inside this rectangle were the mercury delay-line tanks. Many of its novel features later became commonplace, among them alphanumeric as well as numeric processing, magnetic tapes for bulk memory, and circuits called buffers to stop the processor being tied up waiting for data to transfer to tape.[33]

The Univac reflected Eckert's philosophy of loading the vacuum tube circuits conservatively and adding enough redundancy to ensure reliable operation. Statistics gathered by the Metropolitan Life Insurance Company showed that it was available for work 81 percent of the time.[34] The Census Bureau reported, "We never encountered an incorrect solution to a problem which we were sure resulted from an internal computer error."[35]

Univac in Use

On March 31, 1951, Remington Rand turned over the first Univac to the US Census Bureau, beginning commercial sales of large-scale stored program computers in the United States.[36] It wasn't shipped to Washington until late December 1952. After heroic

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efforts to complete and debug the machine, Eckert and Mauchly were apprehensive about dismantling it, moving it, and setting it up again. They needed it nearby to show to other potential customers. By 1954 about twenty systems were built and sold, at around a million dollars each.[37] Table 1.1 lists Univac installations through 1954.

Because Univac followed the general outline of the EDVAC design, it was flexible enough to be used for business applications as well as scientific computations. This meant broadening the sense of what a computer was for. Aiken could not imagine that "the basic logics of a machine designed for the numerical solution of differential equations [could] coincide with the basic logics of a machine intended to make bills for a department store."[38] Eckert and Mauchly knew otherwise. Univac inaugurated the era of large computers for what were later called "data processing" applications.

The closest technology in widespread administrative use was punched card machines, the core product of IBM. The earliest punched card machines, sold from the

Table 1.1
Univac Installations, 1951–1954

#	Date	Customer
1	Summer 1951	US Census Bureau
2	Late 1952	US Air Force, the Pentagon
3	Late 1952	US Army Map Service
4	Fall 1953	US AEC, New York, NY (at NYU)
5	Fall 1953	US AEC, Livermore, CA
6	Fall 1953	David Taylor Model Basin, Carderock, MD
7	1954	Remington Rand, New York, NY
8	1954	General Electric, Louisville, KY
9	1954	Metropolitan Life, New York, NY
10	1954	Wright-Patterson AFB, OH
11	1954	US Steel, Pittsburgh, PA
12	1954	Du Pont, Wilmington, DE
13	1954	US Steel, Gary, IN
14	1954	Franklin Life Insurance, Springfield, OH
15	1954	Westinghouse, Pittsburgh, PA
16	1954	Pacific Mutual Life Insurance, Los Angeles
17	1954	Sylvania Electric, New York, NY
18	1954	Consolidated Edison, New York, NY
19	1954	Consolidated Edison, New York, NY

Note: This list is compiled from a variety of sources and does not include one or two Univacs that were completed but remained with Remington Rand. In some cases the dates are approximate.

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1890s onward to the Census Bureau, were used purely for statistical purposes, tabulating and cross tabulating totals. By the 1930s they were used for many administrative functions and could handle letters as well as numbers and produce printed outputs such as payroll checks. This expansion was reinforced by the deep penetration of IBM's salesforce into the accounting offices of its customers. Punched card machines were often called unit record equipment because a single card encoded information about one thing, such as a sales transaction or employee. A typical small installation consisted of several key punches to get the data onto cards, plus several specialized devices such as tabulators, sorters, and collators. Each machine was configured to carry out the same operation on every card in the deck.

For most customers, what was revolutionary about the Univac was the use of tape in place of punched cards. Univac could scan through a reel of tape, reading selected records, performing some process, and writing the results to another tape.[39] Carrying out all the operations needed for a job meant carrying decks of cards around the room, running them through one machine after another.[40] That made punched card processing labor-intensive. In contrast, the Univac could perform a long sequence of automatic operations before fetching the next record from memory. It replaced not only existing calculating machines but also the people who tended them. Customers regarded the Univac as an information processing system, not a calculator. Published descriptions of the Univac nearly always referred to it as a "tape" machine. For General Electric, for example, "the speed of computing" was "of tertiary importance only."[41]

The Census Bureau, which had been pivotal in getting the fledgling computer company going, hoped to use the Univac for tabulating the 1950 Census. By the time it received its machine in 1951, much of the work had already been put on punched card machines for processing. Then the Bureau had to step aside while the US Air Force and the Atomic Energy Commission commandeered the machine for problems deemed more urgent.[42] Still, Univac #1 was used to cross tabulate census data for four states. Data initially punched onto eleven million cards (one for each person), was transferred to tape for processing by the Univac.[43] The machine was also used for tabulating another subset of the population involving about five million households. Each problem took several months to complete.

Univac #2, installed at the Pentagon for the Air Comptroller in June 1952, was intended for use in Project SCOOP (Scientific Computation of Optimum Problems), which grew out of wartime concerns with getting war materials and men across the Atlantic. The newly created Air Force had mathematically similar challenges in supplying air bases scattered across the globe. Project SCOOP played a key role in the development of linear programming, a widely used technique used to optimize systems involving many variables.[44] Soon after its installation, Univac #2 was put to work on

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SCOOP around the clock.[45] Its uniprinter, based on a Remington Rand electric typewriter, could print only about ten characters per second. That proved a poor match for the high speed tape and processor, but in 1954 Remington Rand delivered the Univac High Speed Printer, able to print a full 130-character line at a time. The Univac installed at Air Force's Air Materiel Command at Wright-Patterson AFB in Ohio performed similar tasks. One of its first jobs was calculating "the complete Fiscal 1956 Budget estimate for airborne equipment space parts, involving approximately 500,000 items."[46] The machine did the job in one day, replacing a battery of punched card equipment.

Some Univacs performed classified weapons work, like the one-of-a-kind computers that preceded them. Univac #5, installed at the Lawrence Livermore Labs in April 1953, was one of those. But even that machine did one unorthodox job. In November 1952, before it was shipped to California, Remington Rand used it to predict Eisenhower's victory over Adlai Stevenson in the 1952 presidential election. Narrated on "live" television, the event inaugurated the intrusion of television into national politics, and of computers into the public's consciousness. For a brief period the word *Univac* was synonymous with *computer*, like *Thermos* for *vacuum bottle*.[47]

A final example of the Univac in use comes from the experience at General Electric's Appliance Park, outside Louisville, Kentucky. Under the direction of Roddy F. Osborn in Louisville and with the advice of the accounting firm Arthur Andersen & Co., General Electric purchased a Univac for four specific tasks: payroll, material scheduling and inventory control, order service and billing, and general cost accounting. These were prosaic operations, already being done by punched card equipment. More advanced applications would commence only after such "bread and butter" work reached a "break even point . . . enough to convince management that a computer system can pay for itself in terms of direct dollar savings (people off the payroll) without waiting for the 'jam' of more glamorous applications."[48]

On Friday, October 15, 1954, the GE Univac first produced payroll checks for the Appliance Park employees.[49] Punched card machines had been doing that job for years, but for an electronic digital computer, which recorded data as invisible magnetic spots on reels of tape, it was a significant milestone. Payroll must be done right and on time. GE had rehearsed the changeover thoroughly and they had arranged with Remington Rand that if their machine broke down and threatened to make the checks late, GE people could bring their tapes to another Univac customer and run the job there.[50] Over the course of the next year they had to exercise this option at least once. On several occasions the checks were printed at the last possible minute, and in the early months it was common to spend much more time doing the job with Univac than it had with punched card equipment. No payrolls were late.

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GE hoped to do more than replace punched card machines. The Louisville plant was conceived as a showcase for modern methods; that was the motivation to locate it in Kentucky rather than Massachusetts or New York, where traditional methods (and labor unions) held sway. Its Univac heralded an age of *automation*, a term coined at the Ford Motor Company in 1947 and popularized by John Diebold in a 1952 book by that title.[51] Diebold spoke of the 1950s as a time when "the push-button age is already obsolete; the buttons now push themselves."[52] Describing the GE installation, Osborn boasted, "While scientists and engineers have been wide-awake in making progress with these remarkable tools, business, like Rip Van Winkle, has been asleep. GE's installation of a Univac may be Rip Van Business's first 'blink.'" He hoped to apply it to long-range planning, market forecasting based on demographic data, revamping production processes to reduce inventories and shipping delays, and similar jobs requiring a more ambitious use of corporate information. At the end of Osborn's essay the *Harvard Business Review*'s editors appended a quotation from Theodore Caplow's *The Sociology of Work*, published that year. It began,

> The Utopia of automatic production is inherently plausible. Indeed, the situation of the United States today, in which poverty has come to mean the absence of status symbols rather than hunger and physical misery, is awesomely favorable when measured against the budgetary experience of previous generations or the contemporary experience of most of the people living on the other continents.[53]

It would not be the last time that the computer would be promoted as the instigator of a digital utopia. Eckert and Mauchly had taken a scientific instrument, similar to a cyclotron, and turned it into a commercial product used for administrative work as well as technical calculation. This was the first transformation of computer technology, but not the last. The "computer age" is really a series of "computer ages." The history of computing is the story of repeated redefinitions of the nature of the computer itself, as it opened new markets, new applications, and new places in the social order. In the mid-1950s, IBM developed a line of products that met the information-handling needs of American businesses. A decade later, military funding turned the computer into a device with which one interacted: a tool to augment one's intellectual capabilities. In the middle of the 1970s, a group of hobbyists and enthusiasts transformed it into a personal appliance. Around 1980, people transformed it from a piece of specialized hardware to a standardized, consumer product defined by its now-commercialized software. Since the 1990s, it has gone through another transformation: into an agent of a world-wide nexus, a communications medium. This process of reinvention and redefinition is still going on.

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